Note on Bandit Algorithms

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1 Bandits, Probability and Concentration

1.1 Introduction

Definition.

- Actions : $A_1, A_2, \cdots \in \mathcal{A}$ and we call $\mathcal{A}$ the action set.
- Rewards : $X_1, X_2, \cdots \in \mathcal{R}$.
- History : $H_{t-1} = (A_1, X_1, \cdots, A_{t-1}, X_{t-1})$.
- Policy : $\pi \in \Pi : H_{t-1} \mapsto A_t$ and we call Π the competitor class.
- Environment: $\mathcal{E} : A_t \mapsto X_t$.
- Regret of a learner relative to a policy π until $n \in \mathbb{N}$ is

$$R_n = \mathbb{E}\left[\sum_{t=1}^{n} \mathcal{E}(\pi(H_{t-1})) - \sum_{t=1}^{n} X_t\right].$$

  where H_0 represents the initial condition.

Example A. n environment is called a **stochastic Bernoulli bandit** if

- $\mathcal{A} = \{1, 2, \ldots, k\}$,
- $X_t \in \{0, 1\}$,
- ${}^{\forall} a \in \mathcal{A}$, $\exists\, \mu_a \in [0,1]^k$ such that $\mathbb{P}[X_t = 1 | A_t = a] = \mu_a$.

Here the optimal policy is to play fixed action $a^* = \underset{a\in\mathcal{A}}{\operatorname{argmax}}\, \mu_a$. That is, the competitor class Π is the set of all constant policies $\Pi = \{\pi_1, \ldots, \pi_k\}$ where π_i chooses action i in every round. The regret relative to the optimal policy π^* until n rounds is

$$\begin{aligned} R_n &= \mathbb{E}\left[\sum_{t=1}^{n} \mathcal{E}(\pi^*(H_{t-1})) - \sum_{t=1}^{n} X_t\right] \\ &= n \max_{a\in\mathcal{A}} \mu_a - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right]. \end{aligned}$$

Definition. Stochastic stationary bandits are the environments restricted to generate the rewards in response to each action from a distribution independent of the previous history.

Example Linear Stochastic Bandits.

- $\mathcal{A} \subset \mathbb{R}^d$,
- $X_t = \langle a, \theta \rangle + \eta_t$ for $\theta \in \mathbb{R}^d, a \in \mathcal{A}$ and η_t is a standard Gaussian.

1.2 Foundations of Probability

Definition. Let an **event set** Ω be given.

- A set $\mathcal{F} \subset 2^\Omega$ is called a **σ-algebra** on Ω if
  (1) $\Omega \in \mathcal{F}$,
  (2) $A \in \mathcal{F} \implies A^c \in \mathcal{F}$,
  (3) $A_1, A_2, \ldots \in \mathcal{F} \implies \bigcup_{i=1}^\infty A_i \in \mathcal{F}$.
- $\mathcal{G} \subseteq \mathcal{F}$ is a **sub-σ-algebra** if $\mathcal{G}$ is a σ-algebra on Ω.
- $(\Omega, \mathcal{F})$ is called a **measurable space** if $\mathcal{F}$ is a σ-algebra on Ω.
- $\mathcal{B}(\mathbb{R})$ is the **Borel σ-algebra** on $\mathbb{R}$, which is the smallest σ-algebra containing all open intervals (a, b) for $a, b \in \mathbb{R}$.

Definition. Let a measurable space $(\Omega, \mathcal{F})$ be given.

- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is **σ-additive** if
  (1) $\mu(\emptyset) = 0$,
  (2) $A_1, A_2, \ldots \in \mathcal{F}$ are disjoint $\implies \mu\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mu(A_i)$.
- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is a **measure** if μ is σ-additive.
- $(\Omega, \mathcal{F}, \mu)$ is called a **measure space** if μ is a measure on $(\Omega, \mathcal{F})$.
- A measure μ is a **probability measure** if $\mu(\Omega) = 1$.
- $(\Omega, \mathcal{F}, \mu)$ is called a **probability space** if μ is a probability measure on $(\Omega, \mathcal{F})$.
- A function $X : \Omega \mapsto \mathbb{R}$ is called a **measurable function** if $^\forall B \in \mathcal{B}(\mathbb{R})$, $X^{-1}(B) \in \mathcal{F}$.
- A **filtration** is a sequence $(\mathcal{F}_t)_{t=0}^n$ of sub-σ-algebras of $\mathcal{F}$ where $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$ for all $t < n$.

Definition. Let a measure space $(\Omega, \mathcal{F}, \mu)$ be given.

- A function $f : \Omega \mapsto [0, \infty]$ is called a **simple function** if $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for some $a_1, a_2, \ldots, a_n \in [0, \infty]$ and $A_1, A_2, \ldots, A_n \in \mathcal{F}$.
- The **integral of a simple function** $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\int_\Omega f d\mu := \sum_{i=1}^n a_i \mu(A_i)$.
- The **integral of a non-negative measurable function** $f : \Omega \mapsto [0, \infty]$ is defined as $\int_\Omega f d\mu := \sup_g \int_\Omega g d\mu$ where the supremum is taken over all simple functions g such that $0 \leq g(\omega) \leq f(\omega)$ for all $\omega \in \Omega$.
- The **integral of a measurable function** $f : \Omega \mapsto \mathbb{R}$ is defined as $\int_\Omega f d\mu := \int_\Omega f^+ d\mu - \int_\Omega f^- d\mu$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.
- A function $f : \Omega \mapsto \mathbb{R}$ is called **integrable** if $\int_\Omega |f| d\mu < \infty$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given.

- A measurable function $X : \Omega \mapsto \mathbb{R}$ is called a **random variable**.
- The **conditional probability** $\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$ for $A, B \in \mathcal{F}$ and $\mathbb{P}[B] > 0$.
- Two events $A, B \in \mathcal{F}$ are called **independent** if $\mathbb{P}[A \cap B] = \mathbb{P}[A] \cdot \mathbb{P}[B]$.
- Two random variables X, Y are called **independent** if $^\forall A, B \in \mathcal{B}(\mathbb{R})$, $\mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A] \cdot \mathbb{P}[Y \in B]$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X be given.

- The **expectation** of a random variable X is defined as $\mathbb{E}[X] := \int_\Omega X d\mathbb{P}$.
- For a sub-σ-algebra $\mathcal{H} \subseteq \mathcal{F}$, the **conditional expectation** of X given $\mathcal{H}$ is denoted by $\mathbb{E}[X|\mathcal{H}]$ and is defined as a random variable g on Ω such that for all $H \in \mathcal{H}$,

$$\int_H g \, d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.11. *If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X is a integrable random variable and $\mathcal{H} \subseteq \mathcal{F}$ is a sub-σ-algebra, there exists a unique (almost surely) random variable $\mathbb{E}[X|\mathcal{H}]$ such that for all $H \in \mathcal{H}$,*

$$\int_H \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.12. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G}, \mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$ be sub-σ-algebras of $\mathcal{F}$, X, Y be integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following properties hold:*

(1) If $X \geq 0$ almost surely, then $\mathbb{E}[X|\mathcal{G}] \geq 0$ almost surely.

(2) $\mathbb{E}[1|\mathcal{G}] = 1$ almost surely.

(3) $\mathbb{E}[X + Y|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}] + \mathbb{E}[Y|\mathcal{G}]$ almost surely.

(4) If Y is $\mathcal{G}$-measurable and $\mathbb{E}[XY] < \infty$, then $\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$ almost surely.

(5) If $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then $\mathbb{E}[X|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1]$ almost surely.

1.3 Stochastic Processes and Markov Chains

Definition. Let two measurable spaces $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be given.

- $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ are called **isomorphic** if there exists a bijective function $f : \mathcal{X} \mapsto \mathcal{Y}$ such that $A \in \mathcal{F} \iff f(A) \in \mathcal{G}$.
- $(\mathcal{X}, \mathcal{F})$ is a **Borel space** if it is isomorphic to $(\mathbb{R}, \mathcal{B}(A))$ for some set $A \in \mathcal{B}(\mathbb{R})$.

Theorem 3.1. *Let μ be a probability measure on a Borel space S and λ be the Lebesgue measure on $([0, 1], \mathcal{B}([0, 1]))$. Then there exists a sequence of independent random element $X_1, X_2, \ldots$ on $([0, 1], \mathcal{B}([0, 1]), \lambda)$ such that $\lambda_{X_t}(\cdot) := \lambda(X_t^{-1}(\cdot)) = \mu(\cdot)$ for all t.*

Definition. For "countable" set $\mathcal{T}$, A stochastic process is a collection of random variables $X_t : t \in \mathcal{T}$.

Theorem 3.2. *Let Borel spaces $(\Omega_n, \mathcal{F}_n)$ and measures μ_n on $(\Omega_1 \times \cdots \times \Omega_n, \mathcal{F}_1 \otimes \cdots \otimes \mathcal{F}_n)$ are given for each $n \in \mathbb{N}$.*

$$\begin{aligned} &{}^\forall n \in \mathbb{N} \text{ and } A \in \mathcal{F}_1 \otimes \cdots \otimes \mathcal{F}_n, \quad \mu_{n+1}(A \times \Omega_{n+1}) = \mu_n(A) \\ &\implies \exists \text{ a probability space } (\Omega, \mathcal{F}, \mathbb{P}) \\ &\text{and random elements } X_1, X_2, \ldots \text{ with } X_t : \Omega \to \Omega_t \text{ such that } \mathbb{P}_{X_1, \ldots, X_n} = \mu_n \text{ for all } n. \end{aligned}$$

Definition. Let $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be measurable spaces. A **probability kernel** or **Markov kernel** form $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y}, \mathcal{G})$ is a function $K : \mathcal{X} \times \mathcal{G} \to [0, 1]$ such that

(a) $K(x, \cdot)$ is a measure for all $x \in \mathcal{X}$ and

(b) $K(\cdot, A)$ is $\mathcal{F}$-measurable for all $A \in \mathcal{G}$.

Notation: $K_x(A)$ or $K(A|x)$

Definition. Let K_1 be a probability kernel from $(\mathcal{X}, \mathcal{F}) \to (\mathcal{Y}, \mathcal{G})$ and K_2 be a probability kernel from $(\mathcal{Y}, \mathcal{G}) \to (\mathcal{Z}, \mathcal{H})$. Then the product kernel $K_1 \otimes K_2$ is the probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y} \times \mathcal{Z}, \mathcal{G} \otimes \mathcal{H})$ defined by

$$(K_1 \otimes K_2)(x, A) = \int_{\mathcal{Y}} \int_{\mathcal{Z}} \mathbb{1}_A((y, z)) K_2(y, dz) K_1(x, dy).$$

Theorem 3.3. *Let $(\Omega_n, \mathcal{F}_n)_{n=1}^{\infty}$ be a sequence of measurable spaces.*

$$\begin{aligned} &K_1 : \textit{a probability measure on } (\Omega_1, \mathcal{F}_1), \\ &K_n : \textit{a probability kernel from } \Pi_{t=1}^{n-1} \Omega_t \textit{ to } \Omega_n \textit{ for } n \geq 2. \\ &\implies \exists \textit{ a probability space } (\Omega, \mathcal{F}, \mathbb{P}) \\ &\textit{and random elements } (X_t)_{t=1}^{\infty} \textit{ with } X_t : \Omega \to \Omega_t \textit{ such that } \mathbb{P}_{X_1, \ldots, X_n} = \bigotimes_{t=1}^{n} K_t \textit{ for all } n. \end{aligned}$$

Definition. A **homogeneous Markov chain** is a sequence of random elements $(X_t)_{t=1}^{\infty}$ taking values in **state space** $\mathcal{S} = (\mathcal{X}, \mathcal{F})$ and with

$$\mathbb{P}(X_{t+1} \in \cdot \,| X_1, \ldots, X_t) = \mathbb{P}(X_{t+1} \in \cdot \,| X_t) = \mu(X_t, \cdot), \qquad \text{a.s.}$$

Here μ is a probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{X}, \mathcal{F})$.

Definition 3.4. A $\mathbb{F}$-adapted sequence of random variables $(X_t)_{t \in \mathbb{N}_+}$(i.e. X_t is $\mathcal{F}_t$-measurable for all $t \in \mathbb{N}$) is a $\mathbb{F}$-adapted **martingale** if

(a) $\mathbb{E}[|X_t|] < \infty$ for all t.

(b) $\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t$ for all t.

Supermartingale if less-than, submartingale if greater-than.

Example 3.5. $(Y_t)_{t \in \mathbb{N}}$ be a sequence of independent random variables with $\mathbb{P}(Y_t = 1) = \mathbb{P}(Y_t = -1) = \frac{1}{2}$. Then the winnings after t rounds is $S_t = \sum_{i=1}^{t} Y_i$, a martingale adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbb{N}_+}$, where $\mathcal{F}_t = \sigma(Y_1, \ldots, Y_t)$.

Definition 3.6. Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration. A random variable $\tau \in \mathbb{N} \cup \{\infty\}$ is a **stopping time** with respect to the filtration $\mathbb{F}$ if

$$\{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}.$$

The σ-algebra generated by the stopping time τ is

$$\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}\}.$$

Theorem 3.8 (Doob's Optional Stopping Theorem)**.** *Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration and $(X_t)_{t \in \mathbb{N}}$ be an $\mathbb{F}$-adapted martingale. Let τ be a stopping time with respect to $\mathbb{F}$, and one of the following conditions is satisfied:*

(a) There exists an $n \in \mathbb{N}$ such that $\mathbb{P}(\tau > n) = 0$.

(b) $\mathbb{E}[\tau] < \infty$ and there exists a constant $c \in \mathbb{R}$ such that $|X_{t+1} - X_t| \leq c$ for all t, $\mathbb{E}[|X_{t+1} - X_t| | \mathcal{F}_t] \leq c$ a.s. for $\tau > t$.

(c) There exists a constant c such that $|X_{t \wedge \tau}| \leq c$ a.s. for all $t \in \mathbb{N}$.

Then X_τ is almost surely well-defined and

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0].$$

Theorem 3.9 (Maximal inequality)**.** *Let $(X_t)_{t=0}^{\infty}$ be a supermartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,*

$$\mathbb{P}\left(\sup_{t \in \mathbb{N}} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_0]}{\epsilon}.$$

Theorem 3.10. *Let $(X_t)_{t=0}^{\infty}$ be a submartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,*

$$\mathbb{P}\left(\max_{t \in 0, 1, \ldots, n} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_n]}{\epsilon}.$$

1.4 Stochastic Bandits

Definition. A **stochastic bandit** is a collection of distributions $\nu = \{P_a : a \in \mathcal{A}\}$.

Here we are challenging to maximize the total reward $S_n = \sum_{t=1}^n X_t$ over n rounds.

Definition. A set of bandits $\mathcal{E}$ for which $\nu \in \mathcal{E}$ is called the environment class.

- An environment class $\mathcal{E}$ is **unstructured** if $\mathcal{A}$ is finite and

$$^{\forall} a \in \mathcal{A}, \exists \text{ sets of distributions } \mathcal{M}_a \text{ such that } \mathcal{E} = \{\nu = (P_a : a \in \mathcal{A}) : P_a \in \mathcal{M}_a, {}^{\forall} a \in \mathcal{A}\}.$$

  or, in short, $\mathcal{E} = \times_{a\in\mathcal{A}} \mathcal{M}_a$.

- Environment classes with finite number of degrees of freedom are called **parametric** (Bernoulli).
- Environment classes that are not parametric are called **nonparametric** (subgaussian).
- An environment class is called **structured** if it is not unstructured.

Example 4.1. Let $\mathcal{A} = \{1, 2\}$ and $\mathcal{E} = \{(\mathcal{B}(\theta), \mathcal{B}(1-\theta)) : \theta \in [0,1]\}$ where $\mathcal{B}(\theta)$ denotes the Bernoulli distribution with mean θ. Then $\mathcal{E}$ is a structured parametric environment class.

Example 4.2 (Stochastic linear bandit)**.** Let $\mathcal{A} \subseteq \mathbb{R}^d$ and $\theta \in \mathbb{R}^d$ and

$$\nu_\theta = (\mathcal{N}(\langle a, \theta \rangle, 1) : a \in \mathcal{A}) \qquad \text{and} \;\; \mathcal{E} = \{\nu_\theta : \theta \in \mathbb{R}^d\}.$$

Example 4.3. Consider an undirected graph G with vertices $V = 1, \ldots, |V|$ and edges $E = 1, \ldots, |E|$. In each round the learner chooses a path from vertex 1 to vertex $|V|$. Then each edge $e \in [E]$ is removed from the graph with probability $1 - \theta_e$ for unknown $\theta \in [0,1]^{|E|}$. In this problem, $\mathcal{A}$ is the set of paths and

$$\nu_\theta = \left(\mathcal{B}\left(\prod_{e\in a} \theta_e\right) : a \in \mathcal{A}\right) \qquad \text{and} \;\; \mathcal{E} = \{\nu_\theta : \theta \in [0,1]^{|E|}\}.$$

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit.

$$\mu_a(\nu) := \int_{-\infty}^{\infty} x \, dP_a(x)$$

Also let $\mu^*(\nu) := \max_{a\in\mathcal{A}} \mu_a(\nu)$, the largest mean of all the arms.

Definition. The **regret** of policy π on bandit instance ν is

$$R_n(\pi, \nu) := n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^n X_t\right]$$

where the expectation is taken with respect to the probability measure on outcomes induced by the interaction of π and ν.

Lemma 4.4. *Let ν be a stochastic bandit environment. Then*

- $R_n(\pi, \nu) \geq 0$ *for all policies* π,
- *the policy* π *choosing* $A_t \in \arg\max_a \mu_a(\nu)$ *for all* t *satisfies* $R_n(\pi, \nu) = 0$, *and*
- *if* $R_n(\pi, \nu) = 0$ *for some policy* π, *then* $\mathbb{P}[\mu_{A_t}(\nu) = \mu^*(\nu)] = 1$ *for all* $t \in [n]$.

Proof. (a) Since $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$,

$$n\mu^*(\nu) \geq \mathbb{E}\left[\sum_{t=1}^{n} \mu_{A_t}(\nu)\right].$$

By the linearity of expectation,

$$\mathbb{E}\left[\sum_{t=1}^{n} \mu_{A_t}(\nu)\right] = \sum_{t=1}^{n} \mathbb{E}\left[\mu_{A_t}(\nu)\right].$$

By the stochastic bandit assumption and the tower property,

$$\begin{aligned}
\sum_{t=1}^{n} \mathbb{E}\left[\mu_{A_t}(\nu)\right] &= \sum_{t=1}^{n} \mathbb{E}\left[\mathbb{E}\left[X_t \mid H_{t-1}, A_t\right]\right] \\
&= \sum_{t=1}^{n} \mathbb{E}\left[X_t\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{n} X_t\right],
\end{aligned}$$

using linearity of expectation again. Therefore,

$$R_n(\pi, \nu) = n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] \geq 0.$$

(b) Let π be the policy choosing $A_t \in \arg\max_a \mu_a(\nu)$ for all t. Then $\mu^*(\nu) = \mu_{A_t}(\nu)$ for all t, which implies

$$\begin{aligned}
R_n(\pi, \nu) &= n\mu^*(\nu) - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] \\
&= \sum_{t=1}^{n} \left(\mu^*(\nu) - \mathbb{E}\left[\mu_{A_t}(\nu)\right]\right) \\
&= \mathbb{E}\left[\sum_{t=1}^{n} \left(\mu^*(\nu) - \mu_{A_t}(\nu)\right)\right] \\
&= 0
\end{aligned}$$

and therefore $R_n(\pi, \nu) = 0$.

(c) Suppose $R_n(\pi, \nu) = 0$ for some policy π. Then

$$\sum_{t=1}^{n} \mathbb{E}\left[\mu^*(\nu) - \mu_{A_t}(\nu)\right] = 0.$$

Since $\mu^*(\nu) - \mu_{A_t}(\nu) \geq 0$ for all t, $\mu^*(\nu) = \mu_{A_t}(\nu)$ almost surely for all $t \in \{1, \ldots, n\}$. $\square$

Remark (Factorisation of the regret)**.** The learner cannot always achieve zero regret, so the relatively weak objective is to find a policy π with

$$\forall\, \nu \in \mathcal{E}, \quad \lim_{n \to \infty} \frac{R_n(\pi, \nu)}{n} = 0.$$

Or further objective, finding some constants $C > 0$ and $p < 1$ such that

$$\forall\, \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \leq Cn^p.$$

Another alternative is to find a function $C : \mathcal{E} \to [0, \infty)$ and $f : \mathbb{N} \to [0, \infty)$ such that

$$^\forall \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \le C(\nu) f(n).$$

Bayesian regret is another candidate objective. For prior probability measure Q on $\mathcal{E}$, the Bayesian regret is the average of the regret with respect to the prior Q, i.e.,

$$\mathrm{BR}_n(\pi, Q) = \int_{\mathcal{E}} R_n(\pi, \nu)\, dQ(\nu)$$

where the regret R_n is measurable.

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit. Then the **suboptimality gap** of action a is defined by

$$\Delta_a(\nu) := \mu^*(\nu) - \mu_a(\nu).$$

Also let

$$T_a(t) := \sum_{s=1}^{t} \mathbb{1}\{A_s = a\}$$

be the number of times action a has been chosen up to time t.

Lemma 4.5 (Regret decomposition lemma)**.** *For any policy π and stochastic bandit ν with $\mathcal{A}$ finite or countable and horizon $n \in \mathbb{N}$, the regret R_n of policy π in ν satisfies*

$$R_n(\pi, \nu) = \sum_{a \in \mathcal{A}} \Delta_a(\nu) \mathbb{E}[T_a(n)].$$

Proof. For any $t \in \mathbb{N}$, $\sum_{a \in \mathcal{A}} \mathbb{1}\{A_t = a\} = 1$. Hence $S_n = \sum_t X_t = \sum_t \sum_a X_t \mathbb{1}\{A_t = a\}$. That is

$$R_n = n\mu^* - \mathbb{E}[S_n] = \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\}].$$

Since $\mathbb{E}[X_t \mid H_{t-1}, A_t] = \mu_{A_t}$, we have

$$\begin{aligned} \mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\} \mid A_t] &= \mathbb{1}\{A_t = a\}\mathbb{E}[\mu^* - X_t \mid A_t] \\ &= \mathbb{1}\{A_t = a\}(\mu^* - \mu_a) \\ &= \mathbb{1}\{A_t = a\}\Delta_a. \end{aligned}$$

Therefore

$$\begin{aligned} R_n &= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\}] \\ &= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[\mathbb{E}[(\mu^* - X_t)\mathbb{1}\{A_t = a\} \mid A_t]] \\ &= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}[\Delta_a \mathbb{1}\{A_t = a\}] \\ &= \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]. \end{aligned}$$

□

1.5 Concentration of measure

Definition. Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables with real values, and assume that the mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{V}[X] < \infty$. Then the **empirical mean** is defined by

$$\hat{\mu} := \frac{1}{n}\sum_{i=1}^{n} X_i.$$

This is an unbiased estimator of the mean μ, since $\mathbb{E}[\hat{\mu}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu$. Using independence of the X_i's, the variance of the empirical mean is

$$\mathbb{V}[\hat{\mu}] = \mathbb{E}\left[(\hat{\mu}-\mu)^2\right] = \frac{\sigma^2}{n}.$$

Two quantities

$$\mathbb{P}[\hat{\mu} \geq \mu + \epsilon] \quad \text{and} \quad \mathbb{P}[\hat{\mu} \leq \mu - \epsilon]$$

are called the **tail probabilities** of $\hat{\mu} - \mu$.

Lemma 5.1. *Let X be a random variable and let $\epsilon > 0$. Then*

(a) (Markov's inequality) : $\mathbb{P}[|X| \geq \epsilon] \leq \frac{\mathbb{E}[|X|]}{\epsilon}$.

(b) (Chebyshev's inequality) : $\mathbb{P}[|X-\mu| \geq \epsilon] \leq \frac{\sigma^2}{\epsilon^2}$.

Proof. (a)

$$\begin{aligned}\mathbb{E}[|X|] &= \int_0^\infty \mathbb{P}[|X| \geq t]\, dt \\ &\geq \int_0^\epsilon \mathbb{P}[|X| \geq \epsilon]\, dt \\ &= \epsilon \mathbb{P}[|X| \geq \epsilon].\end{aligned}$$

(b)

$$\begin{aligned}\mathbb{P}[|X-\mu| \geq \epsilon] &= \mathbb{P}[(X-\mu)^2 \geq \epsilon^2] \\ &\leq \frac{\mathbb{E}[(X-\mu)^2]}{\epsilon^2}.\end{aligned}$$

□

Theorem (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with finite second moment. Define*

$$S_n := \sum_{t=1}^{n}(X_t - \mu).$$

Then for $\sigma^2 > 0$,

$$\frac{S_n}{\sqrt{n}\sigma} \xrightarrow{d} N(0,1).$$

Theorem (Cramér)**.** *Suppose $X_1, \ldots, X_n$ are i.i.d. random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Define μ_n to be the law of S_n. Then μ_n satisfies a large deviation principle on $\mathbb{R}^d$ with rate function given by the Legendre transform of the cumulant generating function of X_1. That is,*

- *For any closed set $F \subset \mathbb{R}^d$,*

$$\limsup_{n\to\infty} \frac{1}{n}\log \mu_n(F) \leq -\inf_{x\in F} I(x),$$

- *For any open set $G \subset \mathbb{R}^d$,*

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(G) \geq - \inf_{x\in G} I(x),$$

where $I(x)$ is the rate function given by the Legendre transform of the cumulant generating function of X_1.

Theorem (Gärtner-Ellis)**.** *Suppose $X_1, \ldots, X_n$ are random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n}\sum_{i=1}^n X_i$. Define μ_n to be the law of S_n. Suppose the limit*

$$\Lambda(\lambda) := \lim_{n\to\infty} \frac{1}{n} \log \mathbb{E}\left[e^{n\langle \lambda, S_n\rangle}\right]$$

exists for all $\lambda \in \mathbb{R}^d$ and $0 \in \mathcal{D}_\Lambda^\circ$, where $\mathcal{D}_\Lambda$ is the effective domain of Λ. Then

- *For any closed set $F \subset \mathbb{R}^d$,*

$$\limsup_{n\to\infty} \frac{1}{n} \log \mu_n(F) \leq - \inf_{x\in F} I(x),$$

- *For any open set $G \subset \mathbb{R}^d$,*

$$\liminf_{n\to\infty} \frac{1}{n} \log \mu_n(G) \geq - \inf_{x\in G\cap\mathcal{F}} \Lambda^*(x),$$

where $\Lambda^*(x)$ is the rate function given by the Legendre transform of $\Lambda(\lambda)$, and $\mathcal{F}$ is the set of exposed points of $\Lambda^*(x)$.

Furthermore, if Λ is essentially smooth and lower semicontinuous, then $\mathcal{F}$ can be replaced by the entire domain of Λ^*.(LDP holds)